

“Warning”

Features:

1. Dual core dual architecture processor: Unique dual core, dual architecture, with a clock speed of 150MHz and strong performance.
2. Large capacity storage: Built in 520KB SRAM and 2MB on-chip Flash to meet various needs.
3. Flexible GPIO design: 15 multifunctional GPIO pins with strong scalability and flexible development.
4. Convenient interface design: Adopting Type-C interface, it can be directly soldered and integrated, and the stamp hole design is convenient for embedded applications.
5. Low power consumption and rich peripherals: Supports low-power mode, accelerated floating-point library, and 12 programmable PIO state machines.

Product Features:

Adopting the official independently designed RP2350A microcontroller chip

Unique dual core and dual architecture, equipped with dual core ARM Cortex-M33 processor and dual core Hazard 3 RISC-V processor, with a clock operating frequency of up to 150MHz

Built in 520KB SRAM and 2MB on-chip Flash

Adopting Type-C interface, keeping up with the trend of the times, no need to worry about forward and backward insertion

Stamp hole design, can be directly welded and integrated into the user's self-designed base plate

USB 1.1 Host and Device Support

Onboard 1 USB expansion port, compatible with USB 2.0/1.1 transmission

Support low-power sleep and sleep mode

Can be recognized as a large capacity storage device through USB for drag and drop downloads

15 GPIO pins of RP2350A lead out

2 SPI, 2 I2C, 2 UART, 4 12 bit ADCs, and 14 controllable PWM channels

Accurate on-chip clock and timer

Built in temperature sensor

On-chip accelerated floating-point library

12 programmable I/O (PIO) state machines for custom peripheral support

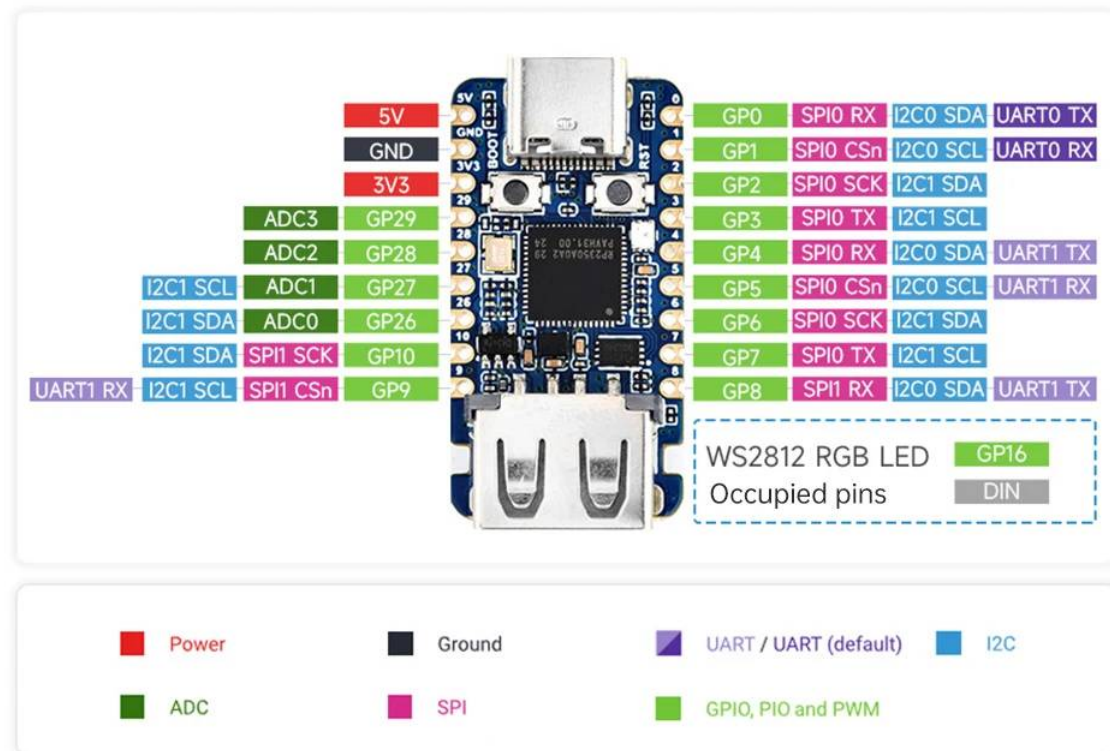
Mini size 33mm * 17.5mm

Package Include□

1X development board

15 multi-function GPIO pins

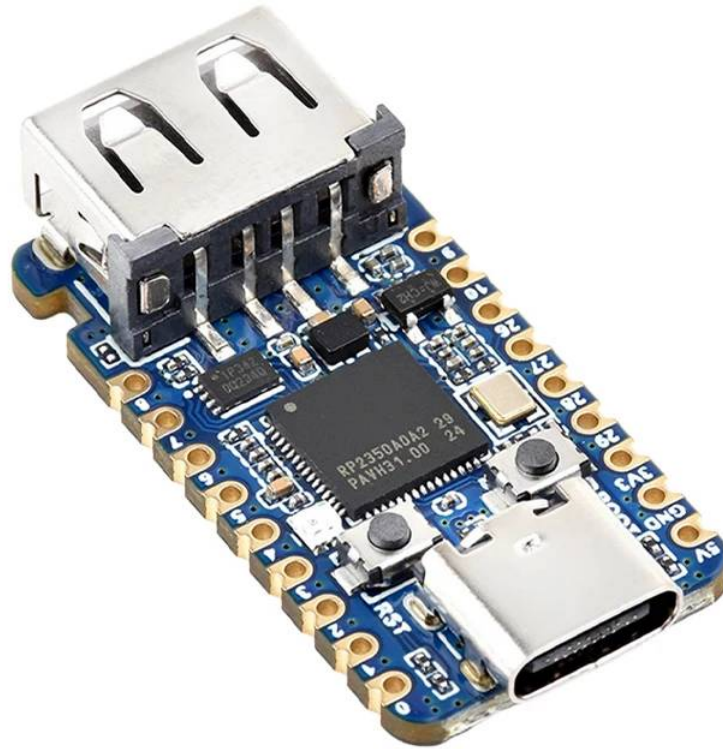
Support for configurable pin functions for easy and flexible development and integration



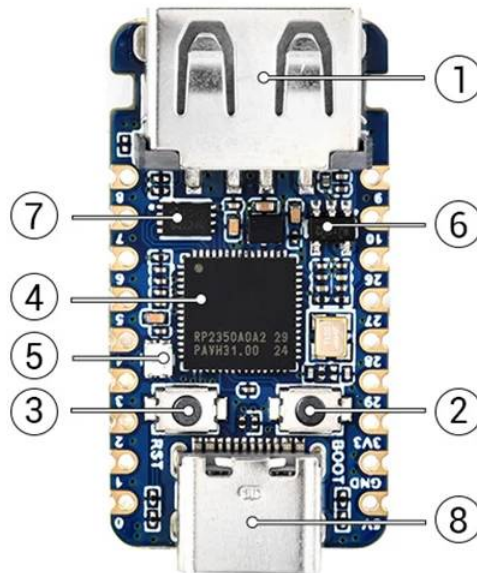
RP2350-USB-A Mini Development Board

Based on RP2350A with on-board USB port

Uses a half-hole process and can be soldered to custom boards



Resource Profile



1. USB 2.0 Type A port

Can be used as a USB master or slave via PIOs

2. BOOT button

Press during reset to enter download mode

3. RESET reset button

4. RP2350A

Dual-core, dual-architecture processor, 150MHz operating frequency

5. WS2812

Cool RGB LED

6. RT9013-33GB

500mA low dropout, low noise, ultra-fast LDO

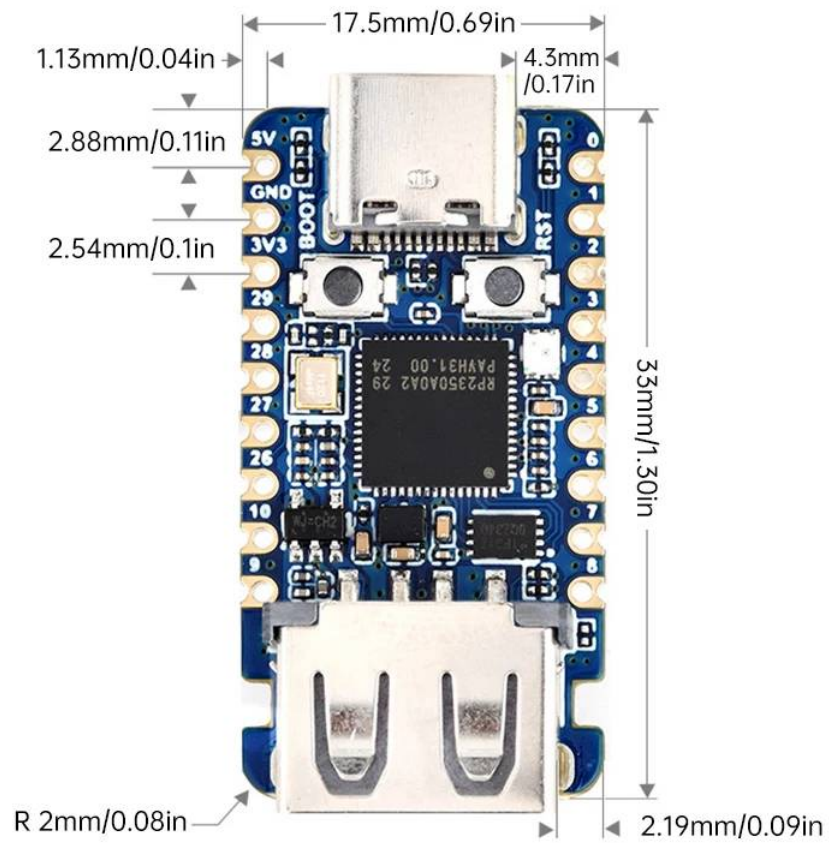
7. W25Q16JVUXIQ

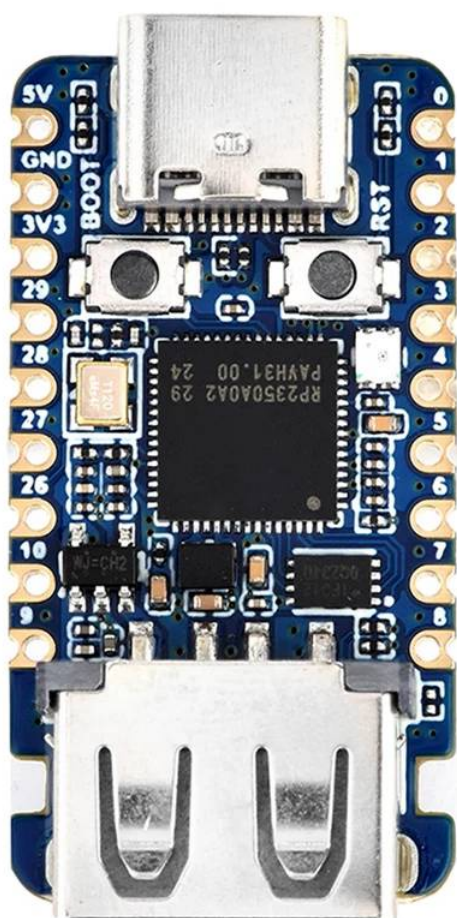
2MB NOR-Flash

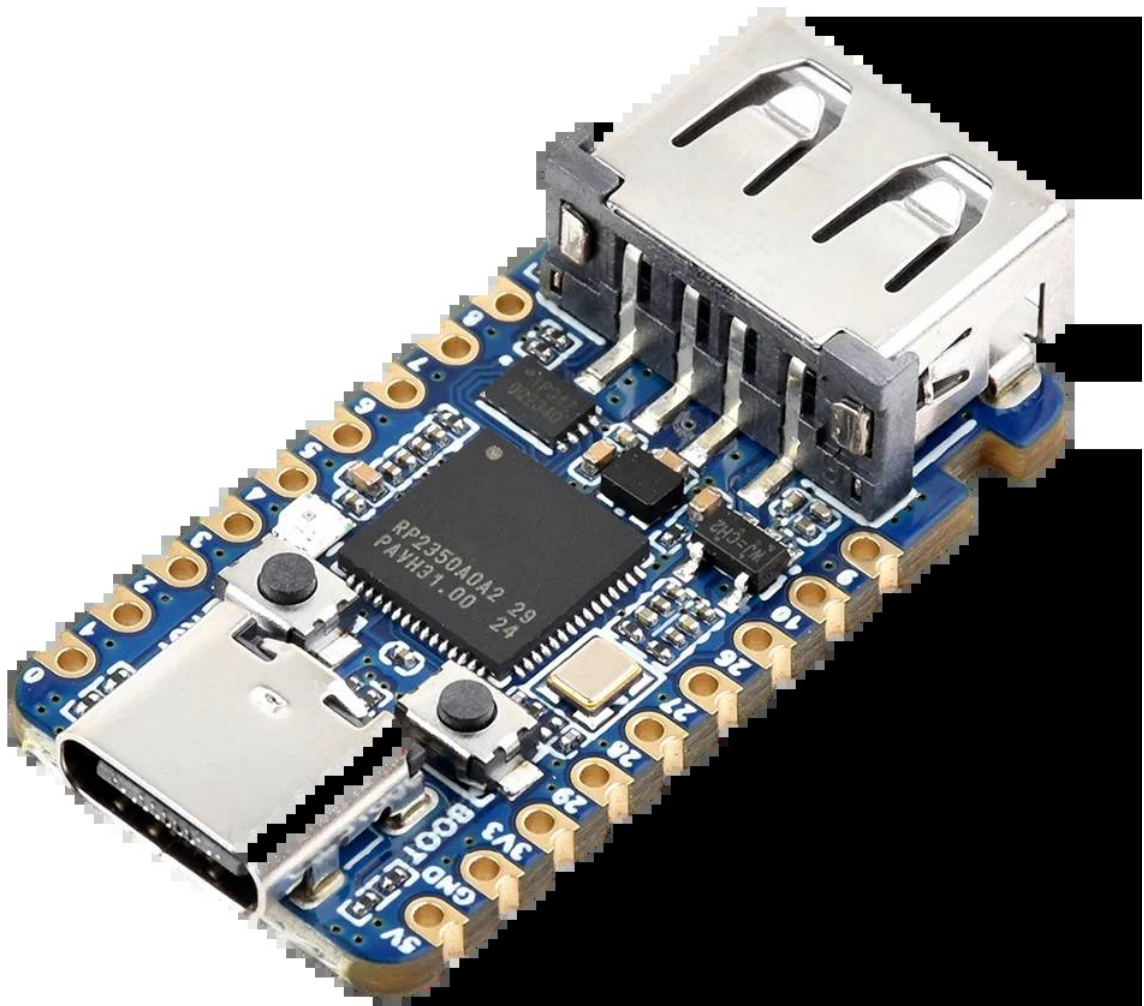
8. USB Type-C port

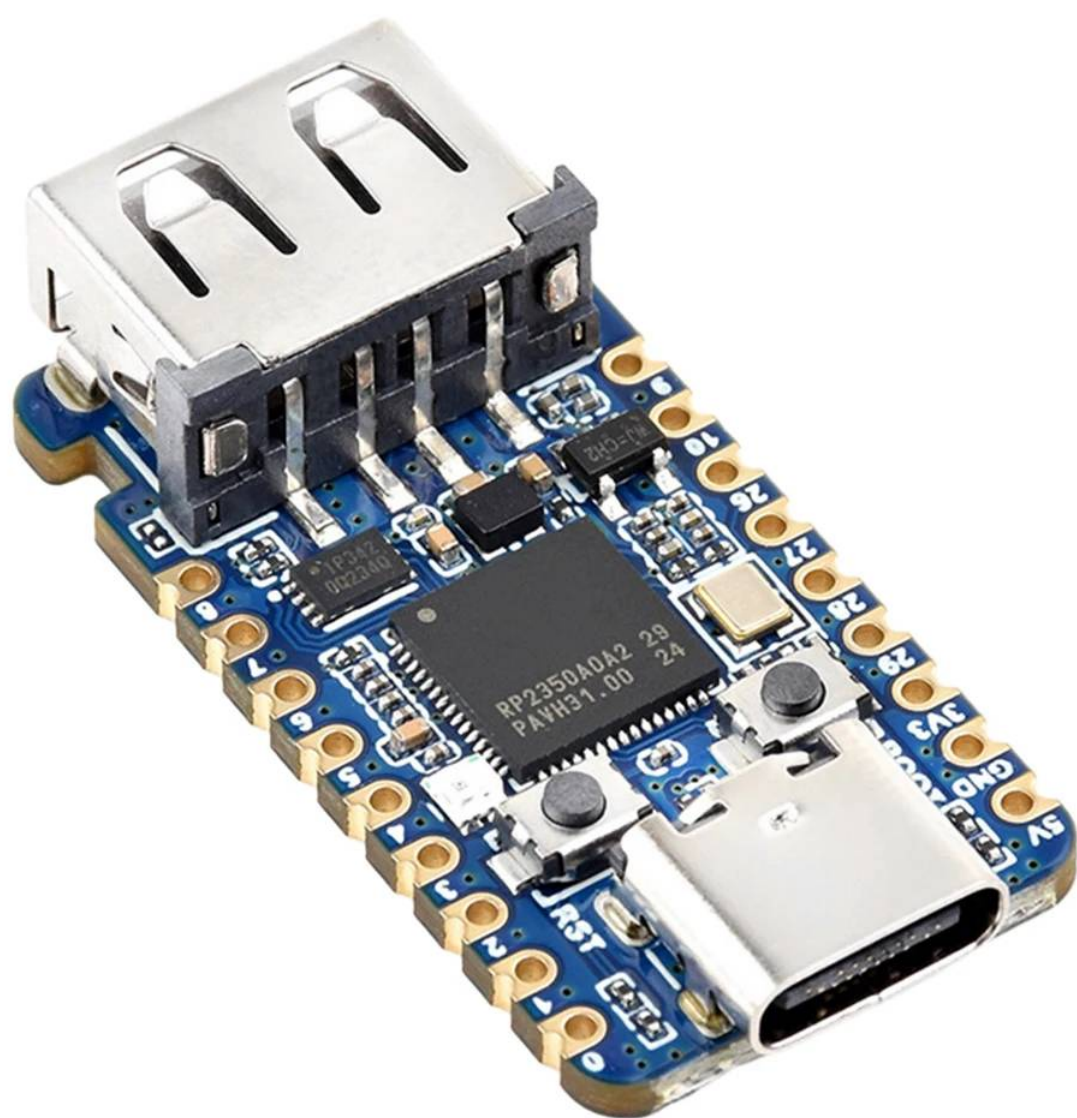
Used for downloading programmes and supports USB 1.1 host and slave devices.

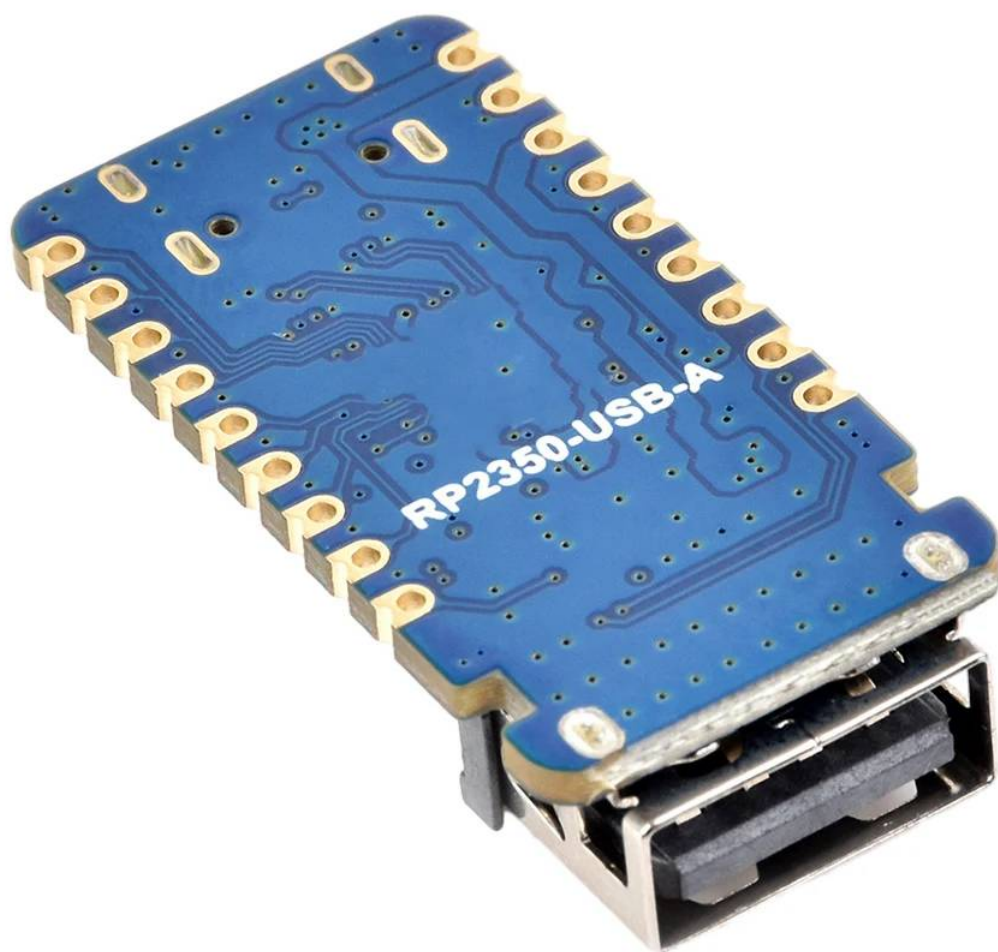
Product Size





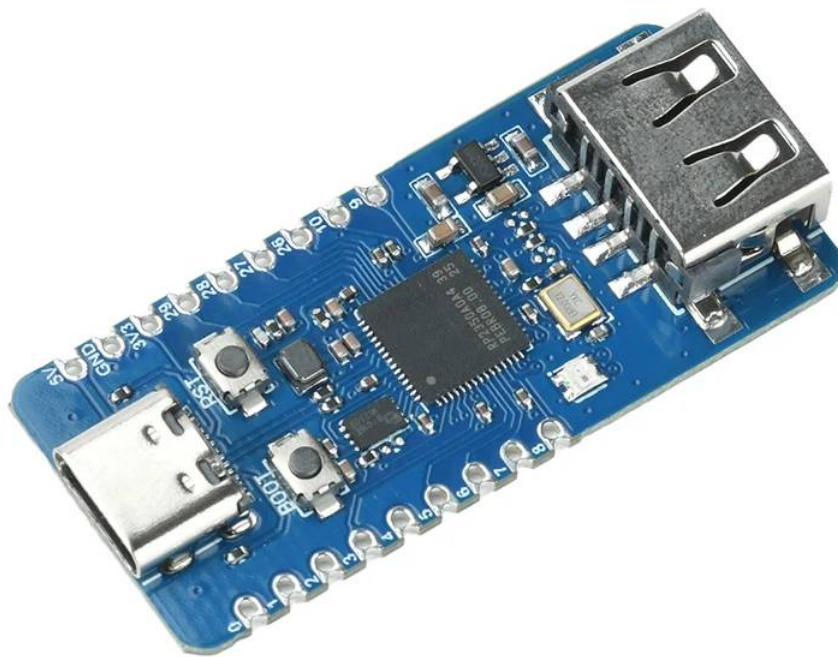






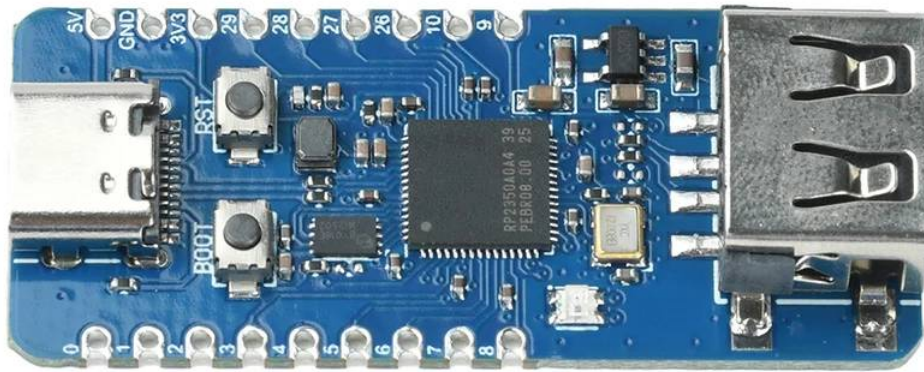
Equipped with RP2350A main controller

Onboard W25Q16 high-capacity Flash



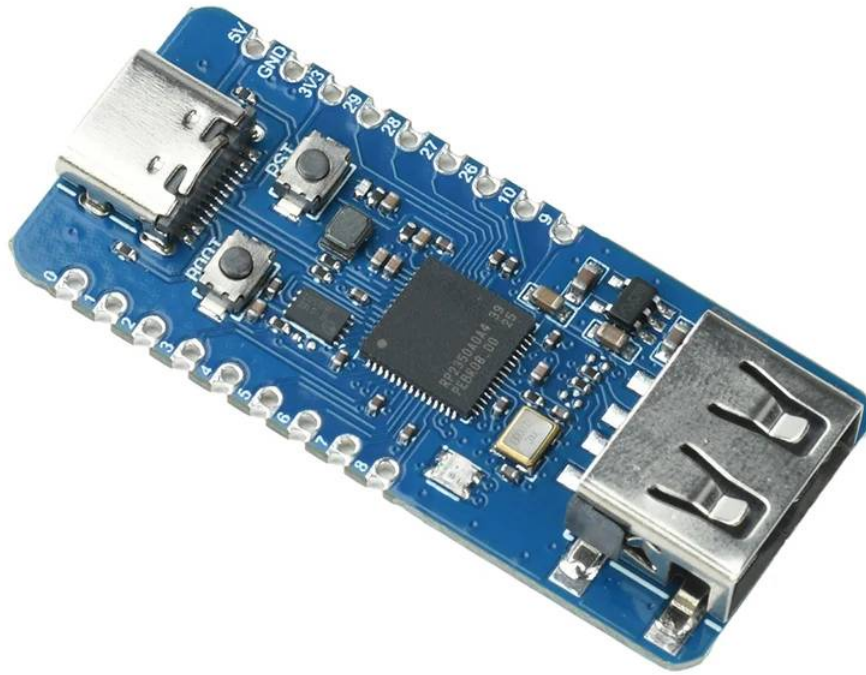
Extract all IO

Equipped with BOOT and RST function buttons



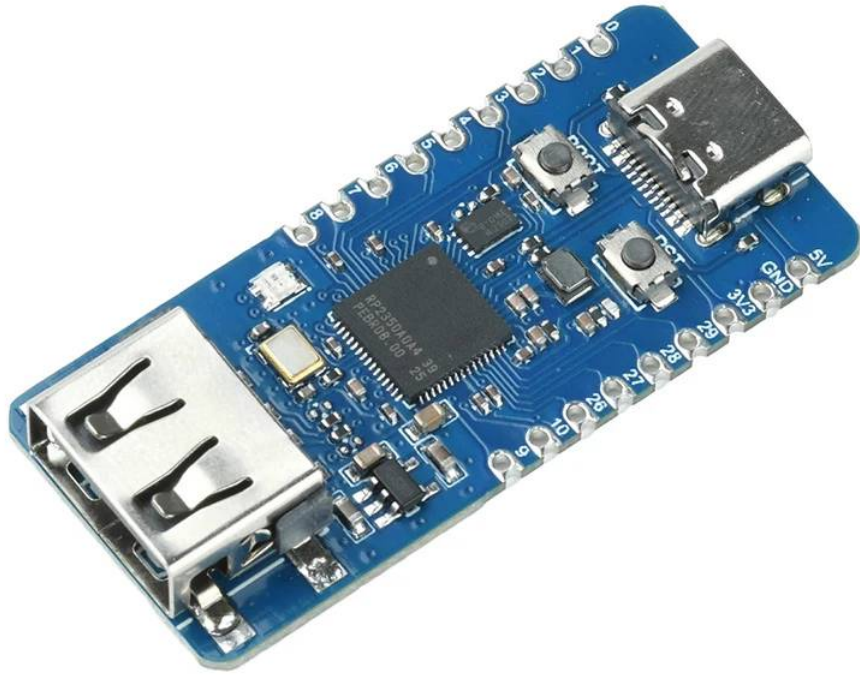
Type-C power supply

Comes with USB-A expansion interface for easy debugging



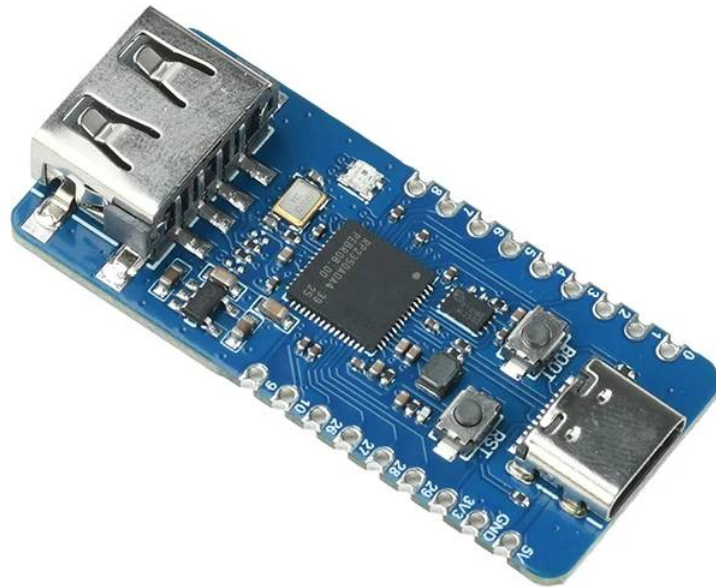
Support multilingual development

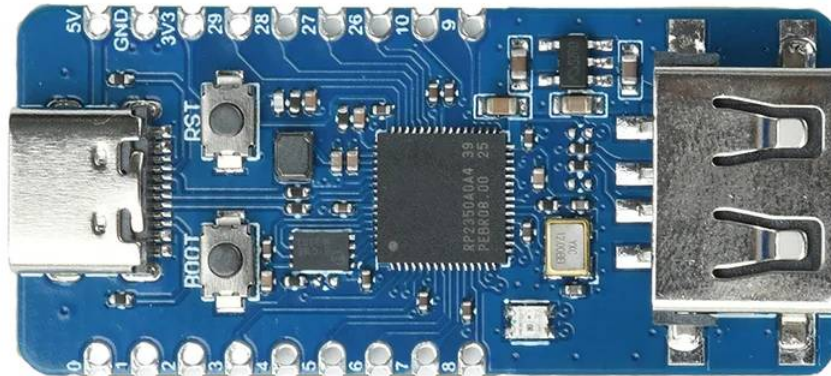
Drag and drop files to quickly burn programs



Half hole miniaturization design

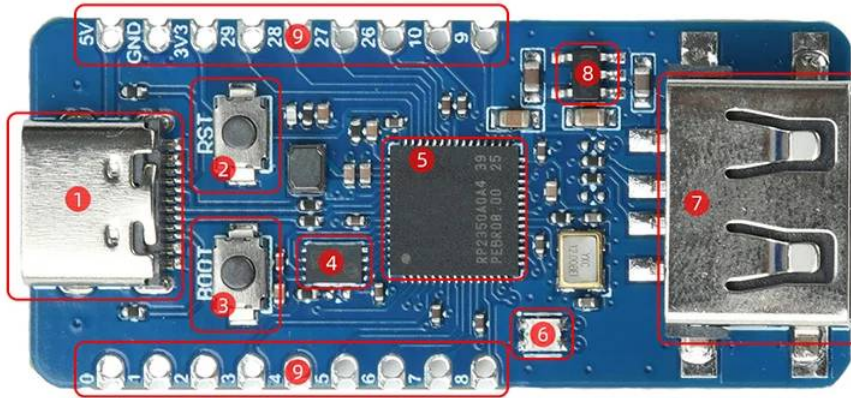
Suitable for the development of embedded control projects



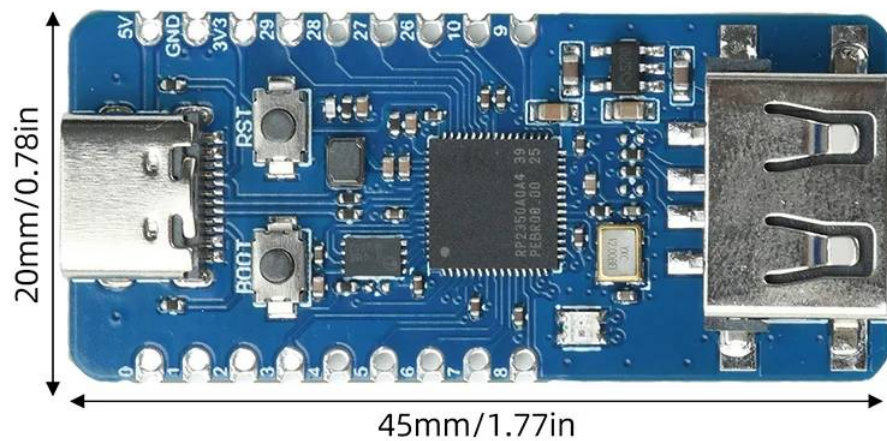


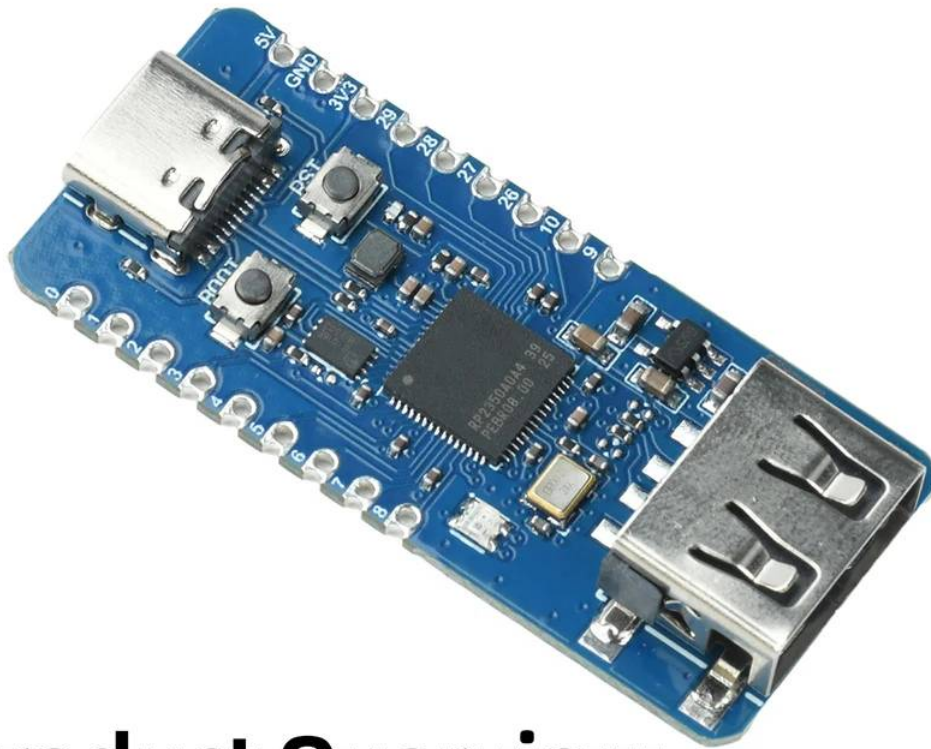
Product Specifications

Interface: Type-C port USB A port	RESET button: Reset button	BOOT button: When resetting Press to enter the burning module
Small size and large capacity FLASH: W25Q16JVUXIQ	MCU: RP2350A (LQFN-60	RGB lights GPIO16
USB-A interface GPIO12,GPIO13	Voltage regulator chip	GPIO Export port
Product size: 45mm * 20mm		



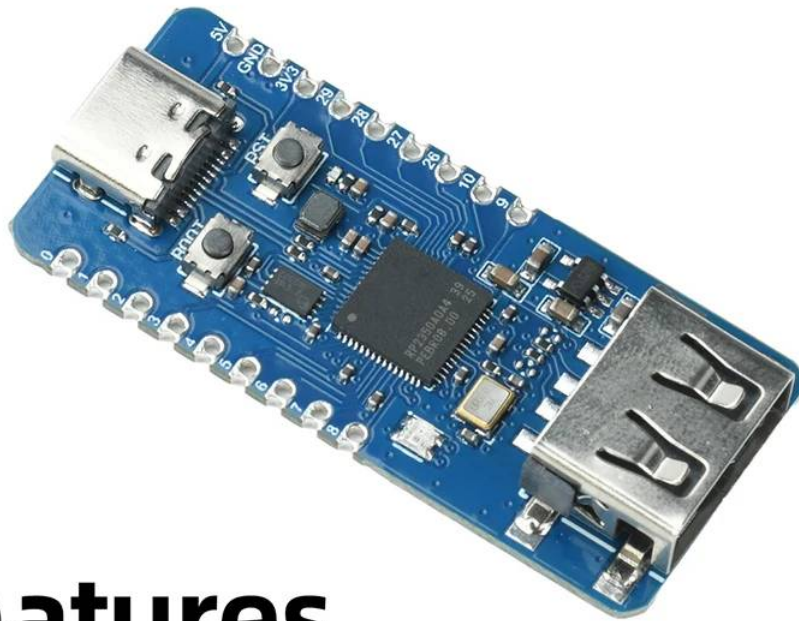
1. Type-C port	4. Small size and large capacity FLASH: W25Q16JVUXIQ	7. USB-A interface GPIO12,GPIO13
2. RESET button: Reset button	5.MCU: RP2350A (LQFN-60)	8. Voltage reduction and stabilization chip
3. BOOT button: Press to reset Enter the burning module	6. RGB lights GPIO16	9.GPIO Export port





Product Overview

This RP2350_V2 USB expansion development board is developed based on RP2350A, with a small-sized FLASH-W25Q16JVUXIQ onboard. It uses a Type-C port, an IO port, and onboard BOOT and RST keys. There is also one USB expansion port onboard for easy user debugging. It supports various development environments such as C, C++, Micro Python, Ard, etc. The module can be recognized as a large capacity storage through USB for program burning.



Features

IO port output and onboard BOOT and RST keys

FLASH-W25Q16JVUXIQ with small size and large capacity onboard

Supports multiple development environments such as C, C++, Micro Python, Ard, etc

Designed with half hole technology, it can be embedded and combined into various products

Support computer USB recognition, copy the program into a large capacity storage device to achieve burning

Onboard 1 USB expansion port

Using a serial debugging assistant (such as ATK_XCOM), open the virtual serial port generated by RP2350 with a baud rate of 115200. When moving the mouse or typing the keyboard, the serial port will continuously print out the captured data packets (such as coordinate changes or threshold presses)



USB testing

1: After holding down the BOOT button, connect the module to the computer using a Type-C cable. The computer will recognize a removable disk and copy the uf2 file into it to complete the firmware burning.



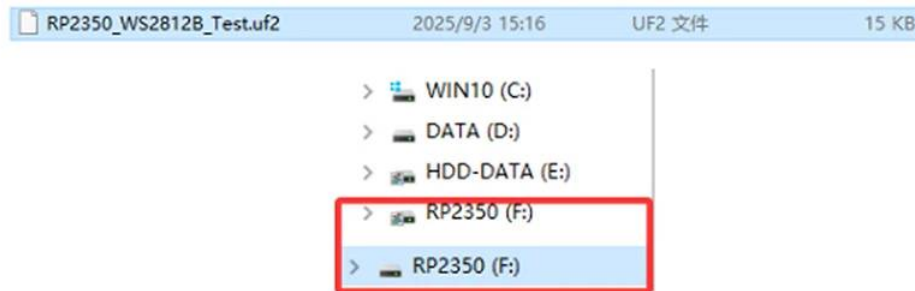
2: First, connect the mouse/keyboard to port A of the RP2350 development board, and then connect port C of the RP2350 to the computer



Instructions for Use

1. RGB lamp test

1: After holding down the BOOT button, connect the module to the computer using a Type-C cable. The computer will recognize a removable disk and copy the .uf2 file to this disk to complete firmware burning



2: After burning. RGB lights light up in different colors

